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Statistics for Business Administration

Exam: MA9712 / Practice Exam A – Solutions **Date:** Monday 27th July, 2026
Examiner: Prof. Mathias Drton **Time:** 10:00 – 11:30

Working instructions

- This exam consists of pages with a total of 9 problems.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is **52 credits**.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one scientific pocket calculator
 - the cheat sheet, which is contained in this exam on page 16
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____

Early submission at _____

Problem 1

7.5 credits

A technology firm surveys a random sample of 480 of its remote employees about a proposed switch to a four-day work week. Each employee is recorded as being *For*, *Against*, or *Undecided*, and is cross-tabulated by age group (Under 30, 30–45, Over 45). The results are:

	For	Against	Undecided	Total
Under 30	110	30	20	160
30–45	80	60	40	180
Over 45	50	54	36	140
Total	240	144	96	480

- a) Identify the response variable and the explanatory variable in this study, and state whether each is categorical or numerical.

Solution. The **response** variable is the opinion about the four-day work week (For/Against/Undecided): categorical. The **explanatory** variable is the age group (Under 30 / 30–45 / Over 45): categorical (ordinal).

- b) The employees were selected by a simple random sample from all remote employees of this firm. Can the results be generalized to all remote employees of the firm? Can we conclude that age *causes* a difference in opinion about the four-day work week? Justify both answers.

Solution. Generalization: yes. Because the sample is a simple random sample from all remote employees, the results generalize to that population. **Causation: no.** This is an observational study (no random assignment of age), so an association does not imply that age causes the difference in opinion; confounding variables may be present.

- c) Would a scatterplot be an appropriate display for these two variables? Justify your answer.

Solution. No. A scatterplot displays the relationship between two *numerical* variables. Here both variables are categorical, so an appropriate display is a (segmented/side-by-side) bar chart or a mosaic plot, not a scatterplot.

- d) An employee is chosen at random. Compute the probability that the employee is in the 30–45 age group *and* is For the change.

Solution.

$$\mathbb{P}(30\text{--}45 \cap \text{For}) = \frac{80}{480} = 0.1\bar{6} \approx 0.167.$$

$$\mathbb{P}(30\text{--}45 \cap \text{For}) = 80/480 \approx 0.167$$

- e) Show that $\mathbb{P}(\text{For} \mid \text{Under 30}) = 0.688$ (rounded to three decimal places).

Solution.

$$\mathbb{P}(\text{For} \mid \text{Under 30}) = \frac{\mathbb{P}(\text{For} \cap \text{Under 30})}{\mathbb{P}(\text{Under 30})} = \frac{110/480}{160/480} = \frac{110}{160} = 0.6875 \approx 0.688.$$

- f) Suppose 10 employees are selected independently at random, and let X be the number who are Against the change. Assuming $\mathbb{P}(\text{Against}) = 0.30$, compute $\mathbb{P}(X = 4)$. The following hint is provided:

```
| dbinom(4, size=10, prob=0.30) ## [1] 0.2001
| pbinom(3, size=10, prob=0.30) ## [1] 0.6496
```

Solution. $X \sim \text{Bin}(10, 0.30)$, so

$$\mathbb{P}(X = 4) = \binom{10}{4} (0.30)^4 (0.70)^6 = 0.2001.$$

$$\mathbb{P}(X = 4) \approx 0.2001$$

Problem 2

6 credits

A café owner records the morning wait time (in minutes) for a random sample of $n = 64$ mornings. The sample mean is $\bar{x} = 4.5$ minutes and the sample standard deviation is $s = 1.6$ minutes. An empirical cumulative distribution function (ECDF) of the wait times shows that 25% of mornings had a wait time of 3.6 minutes or less, and 75% had a wait time of 5.4 minutes or less.

- a) Read off the first quartile Q_1 and third quartile Q_3 from the ECDF description and compute the interquartile range (IQR).

Solution. $Q_1 = 3.6$ min (ECDF reaches 0.25), $Q_3 = 5.4$ min (ECDF reaches 0.75). Then $\text{IQR} = Q_3 - Q_1 = 5.4 - 3.6 = 1.8$ minutes.

- b) State the conditions required for the sampling distribution of $\bar{x}$ to be approximately normal (Central Limit Theorem), and argue whether they are satisfied here.

Solution. Conditions: (i) **independence** — observations are independent, met by random sampling of mornings (and $n = 64$ is well below 10% of all possible mornings); (ii) **sample size / normality** — with $n = 64 \geq 30$ the CLT ensures $\bar{x}$ is approximately normal even if the population is skewed. Both conditions are satisfied.

- c) Construct a 95% confidence interval for the true mean morning wait time. Use the hint:

```
| qnorm(0.975)    ## [1] 1.96
```

Solution.

$$\bar{x} \pm z^* \frac{s}{\sqrt{n}} = 4.5 \pm 1.96 \cdot \frac{1.6}{\sqrt{64}} = 4.5 \pm 1.96 \cdot 0.2 = 4.5 \pm 0.392.$$

[4.108, 4.892] minutes

- d) Interpret the confidence interval from part c) in the context of this study.

Solution. We are 95% confident that the true mean morning wait time is between 4.11 and 4.89 minutes.

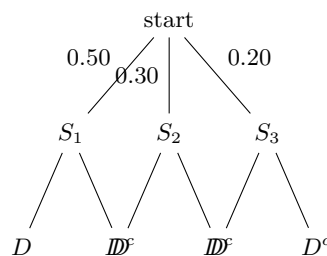
Problem 3

4 credits

A factory assembles a product using a single electronic component, which it buys from three suppliers. Supplier 1 provides 50% of the components, Supplier 2 provides 30%, and Supplier 3 provides 20%. The proportion of defective components is 2% for Supplier 1, 4% for Supplier 2, and 5% for Supplier 3.

- a) Define appropriate events and describe (in words or with a labelled tree) the probability model for “supplier of a randomly chosen component” and “component is defective”.

Solution. Let S_1, S_2, S_3 be the events that a component comes from Supplier 1, 2, 3 respectively, and let D be the event that the component is defective. Then $\mathbb{P}(S_1) = 0.50$, $\mathbb{P}(S_2) = 0.30$, $\mathbb{P}(S_3) = 0.20$ and $\mathbb{P}(D | S_1) = 0.02$, $\mathbb{P}(D | S_2) = 0.04$, $\mathbb{P}(D | S_3) = 0.05$. The tree branches first on supplier, then on defective/non-defective.



- b) A randomly chosen component is found to be defective. Compute the probability that it came from Supplier 3. Give your answer to four decimal places.

Solution. By the law of total probability,

$$\mathbb{P}(D) = \sum_i \mathbb{P}(S_i) \mathbb{P}(D | S_i) = 0.50(0.02) + 0.30(0.04) + 0.20(0.05) = 0.010 + 0.012 + 0.010 = 0.032.$$

By Bayes' theorem,

$$\mathbb{P}(S_3 | D) = \frac{\mathbb{P}(S_3) \mathbb{P}(D | S_3)}{\mathbb{P}(D)} = \frac{0.20 \cdot 0.05}{0.032} = \frac{0.010}{0.032} = 0.3125.$$

$\mathbb{P}(S_3 | D) = 0.3125$

Problem 4

5 credits

A bakery advertises that at most 3% of its loaves are underweight. A consumer-protection agency takes a random sample of $n = 150$ loaves and finds that 9 of them are underweight. Test, at significance level $\alpha = 0.05$, whether the data provide convincing evidence that the true proportion of underweight loaves exceeds 3%.

- a) State the null and alternative hypotheses in terms of the population proportion p .

Solution. $H_0 : p = 0.03$ versus $H_A : p > 0.03$ (one-sided).

- b) Let X be the number of underweight loaves among the 150. State the distribution of X under H_0 , and identify which values of X count as “at least as extreme” as the observed value.

Solution. Under H_0 , $X \sim \text{Bin}(150, 0.03)$. Because H_A is one-sided to the right, “at least as extreme” as the observed $X = 9$ means $X \geq 9$.

- c) Carry out the exact binomial test and state your conclusion in context, using the hint:

```
| pbinom(8, size=150, prob=0.03) ## [1] 0.9622
```

Solution.

$$\text{p-value} = \mathbb{P}(X \geq 9) = 1 - \mathbb{P}(X \leq 8) = 1 - \text{pbinom}(8, 150, 0.03) = 1 - 0.9622 = 0.0378.$$

p-value = 0.0378 Since the p-value 0.0378 is less than the significance level 0.05, we reject H_0 . The data provide convincing evidence that more than 3% of the bakery's loaves are underweight.

Problem 5

4 credits

Decide for each of the following statements whether it is **true** or **false**, and *justify* your decision.

Remark: Only correct justifications are awarded with points, not the decision (true/false).

- a) In a sample of $n = 40$ observations with sample proportion $\hat{p} = 0.05$, the success–failure condition is satisfied, so a normal approximation for the sampling distribution of $\hat{p}$ is appropriate.

Solution. False. The success–failure condition requires $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$. Here $n\hat{p} = 40(0.05) = 2 < 10$, so the condition fails and the normal approximation is *not* appropriate.

- b) In an observational study drawn as a simple random sample from a population, a statistically significant association between two variables allows us to conclude that one variable causes the other.

Solution. False. Random sampling supports *generalization* to the population but not causation. Only a randomized experiment (random assignment of treatments) allows a causal conclusion; in an observational study confounding variables may explain the association.

- c) A p-value of 0.03 means that there is a 3% probability that the null hypothesis is true.

Solution. False. The p-value is the probability of observing data at least as extreme as the observed data *assuming* H_0 is true. It is not the probability that H_0 is true.

- d) If a 95% confidence interval for a mean is $[10, 14]$, then the corresponding 90% confidence interval (same data) is contained within $[10, 14]$.

Solution. True. A lower confidence level uses a smaller critical value ($z_{0.95} = 1.645 < 1.96 = z_{0.975}$), hence a smaller margin of error. With the same center $\bar{x} = 12$, the 90% interval is narrower and therefore contained within the 95% interval $[10, 14]$.

Problem 6

4 credits

A university surveys a random sample of 300 students and records their field of study (Business, Engineering, Sciences) and their preferred learning mode (In-person, Hybrid, Online). We wish to test whether field of study and preferred learning mode are independent. The observed counts are:

	In-person	Hybrid	Online	Total
Business	40	50	30	120
Engineering	30	40	20	90
Sciences	30	30	30	90
Total	100	120	80	300

- a) State the null and alternative hypotheses.

Solution. H_0 : field of study and preferred learning mode are *independent*. H_A : field of study and preferred learning mode are *not independent* (associated).

- b) Compute the expected count for the (Business, Hybrid) cell and verify that the condition on expected counts for the chi-square test is met.

Solution.

$$E_{\text{Bus,Hyb}} = \frac{(\text{row total})(\text{col total})}{\text{grand total}} = \frac{120 \cdot 120}{300} = 48.$$

The smallest expected count in the table is $E_{\text{Eng,Online}} = 90 \cdot 80 / 300 = 24 \geq 5$; since all expected counts are ≥ 5 , the chi-square condition is satisfied.

- c) The test statistic is $\chi^2 = 3.819$. State the degrees of freedom and bracket the p-value, then conclude at $\alpha = 0.05$. Use the hint:

```
qchisq(c(0.90,0.95,0.975,0.99), df=4)
## [1]  7.779  9.488 11.143 13.277
```

Solution. Degrees of freedom: $df = (r - 1)(c - 1) = (3 - 1)(3 - 1) = 4$. Since $\chi^2 = 3.819 < 7.779 = q_{0.90}$, the upper-tail probability satisfies p-value > 0.10 . p-value > 0.10 Since the p-value is larger than 0.05, we fail to reject H_0 . The data do not provide convincing evidence that field of study and preferred learning mode are associated.

Problem 7

9 credits

An agronomist compares three fertilizers (A, B, C) for their effect on crop yield (in kg per plot). Each fertilizer is applied to 6 plots, for a total of $n = 18$ plots. A group summary is:

```
# A tibble: 3 x 4
  fertilizer mean    sd     n
  <chr>      <dbl> <dbl> <int>
1 A          42.0   3.0     6
2 B          45.0   4.0     6
3 C          51.0   5.0     6
```

- a) State the null and alternative hypotheses for the ANOVA F-test.

Solution. $H_0 : \mu_A = \mu_B = \mu_C$ (all three mean yields are equal). H_A : at least one fertilizer's mean yield differs from the others.

- b) State the three conditions required for ANOVA and comment briefly on whether each is plausible here.

Solution. (i) **Independence** — plots and groups are independent (plausible if plots are randomly assigned/separated). (ii) **Normality** — yields within each group are approximately normal (each $n_i = 6$ is small, so this should be checked, e.g. via normal quantile plots). (iii) **Constant variance** — group SDs are 3, 4, 5; the largest (5) is less than twice the smallest (3), so roughly equal variances are plausible.

- c) Fill in the missing entries (marked NA) of the ANOVA table below. The grand mean of all 18 yields is $\bar{y} = 46$.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fertilizer	NA	NA	NA	NA	NA
Residuals	NA	250	NA		

Solution.

$$\begin{aligned}
 \text{SSG} &= \sum_i n_i (\bar{y}_i - \bar{y})^2 = 6[(42 - 46)^2 + (45 - 46)^2 + (51 - 46)^2] \\
 &= 6(16 + 1 + 25) = 6 \cdot 42 = 252, \\
 df_{\text{group}} &= k - 1 = 2, \quad df_{\text{resid}} = n - k = 18 - 3 = 15, \\
 \text{MSG} &= \frac{252}{2} = 126, \quad \text{MSE} = \frac{250}{15} = 16.667, \quad F = \frac{126}{16.667} = 7.56.
 \end{aligned}$$

Completed table:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fertilizer	2	252	126.000	7.560	0.0054
Residuals	15	250	16.667		

- d) Conduct the F-test at $\alpha = 0.05$ and state your conclusion in context, using the hint:

```
| qf(c(0.95,0.99), df1=2, df2=15) ## [1] 3.682 6.359
```

Solution. Since $F = 7.56 > 6.359 = q_{0.99}$, we have $p\text{-value} < 0.01 < 0.05$, so we reject H_0 . The data provide convincing evidence that at least one fertilizer differs in mean crop yield.

- e) What percentage of the variability in crop yield is explained by fertilizer? (i.e. report $R^2 = \text{SSG}/\text{SST}$.)

Solution.

$$\text{SST} = \text{SSG} + \text{SSE} = 252 + 250 = 502, \quad R^2 = \frac{\text{SSG}}{\text{SST}} = \frac{252}{502} = 0.5020 \approx 50.2\%.$$

$$R^2 \approx 0.502 \text{ (50.2\%)}$$

- f) If all three pairwise comparisons are to be carried out afterwards using a Bonferroni correction with a family-wise error rate of 0.05, what significance level should be used for each individual pairwise test?

Solution. Number of pairwise comparisons $K = \binom{3}{2} = 3$, so

$$\alpha^* = \frac{\alpha}{K} = \frac{0.05}{3} = 0.0167.$$

$$\alpha^* = 0.05/3 \approx 0.0167$$

Problem 8

8.5 credits

A real-estate analyst fits a multiple linear regression model for the sale price of a house (in 1000 EUR) on its size (in m²), number of bedrooms, age (in years), and distance to the city center (in km), using data on $n = 100$ houses. The fitted model output is:

```
Call:
lm(formula = price ~ size + bedrooms + age + distance)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  150.000    20.500    7.317  8.0e-11 ***
size          2.500     0.300    8.333  5.9e-13 ***
bedrooms     12.000     4.000    3.000  0.00345 **
age          -1.500     0.450   -3.333  0.00122 **
distance     -8.000     2.100   -3.810  0.00025 ***
---
Residual standard error: 23.94 on 95 degrees of freedom
Multiple R-squared:  0.78
F-statistic: 84.2 on 4 and 95 DF,  p-value: < 2.2e-16
```

- a) Write down the fitted regression equation.

Solution.

$$\widehat{\text{price}} = 150 + 2.5 \text{ size} + 12 \text{ bedrooms} - 1.5 \text{ age} - 8 \text{ distance}.$$

- b) Interpret the estimated slope coefficients for **size** and for **distance** in context.

Solution. size: All else equal, each additional square meter of size is associated with an increase in the predicted price of 2.5 thousand EUR (i.e. 2500 EUR). **distance:** All else equal, each additional kilometer from the city center is associated with a decrease in the predicted price of 8 thousand EUR (i.e. 8000 EUR).

- c) State the null and alternative hypotheses for testing whether **bedrooms** has an effect on price (given the other predictors), and state the test decision at $\alpha = 0.05$.

Solution. $H_0 : \beta_{\text{bedrooms}} = 0$ versus $H_A : \beta_{\text{bedrooms}} \neq 0$. The reported p-value is $0.00345 < 0.05$, so we reject H_0 : given the other predictors, the number of bedrooms has a statistically significant effect on price.

- d) The sample variance of **price** is $\text{Var}(\text{price}) = 2500$ (in 1000-EUR²). Using the reported $R^2 = 0.78$, compute the adjusted R^2 .

Solution. With $n = 100$, $k = 4$,

$$R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n-1}{n-k-1} = 1 - (1 - 0.78) \frac{99}{95} = 1 - 0.22 \cdot 1.0421 = 1 - 0.2293 = 0.7707.$$

$R_{\text{adj}}^2 \approx 0.771$ (Equivalently, $\text{SST} = \text{Var}(\text{price}) \cdot (n-1) = 2500 \cdot 99 = 247500$, $\text{RSS} = (1 - 0.78) \cdot 247500 = 54450$, giving the same R_{adj}^2 .)

- e) In a backward elimination based on AIC, which single predictor is the most likely candidate to be dropped first, and why?

Solution. bedrooms. It has the largest p-value (0.00345) equivalently the smallest $|t|$ -value (3.000) among the predictors, so it contributes the least and its removal would most reduce (or least increase) the AIC. All predictors are significant here, so no variable is actually dropped, but **bedrooms** is the first candidate considered.

Problem 9

5 credits

A bank models whether a customer defaults on a loan (**default**: yes/no) using the customer's annual income (in 1000 EUR), credit score, and loan amount (in 1000 EUR). A logistic regression is fitted on 300 customers:

```
glm(formula = default ~ income + credit_score + loan_amount,
    family = _____ )
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.20000	1.10000	4.727	2.3e-06 ***
income	-0.03000	0.01000	-3.000	0.00270 **
credit_score	-0.00800	0.00250	-3.200	0.00137 **
loan_amount	0.04500	0.01500	3.000	0.00270 **

- a) Fill in the blank in the `family = _____` argument so that this is a logistic regression.

Solution. `family = binomial` (equivalently `binomial(link = "logit")`).

- b) Interpret the `credit_score` coefficient as an odds ratio for a 10-point increase in credit score.

Solution. For a 10-point increase, the log-odds change by $10 \cdot (-0.008) = -0.08$, so

$$\text{OR} = e^{10 \cdot (-0.008)} = e^{-0.08} = 0.9231.$$

The odds of default is decreased by $(1 - 0.9231) \cdot 100\% = 7.69\%$ for each 10-point increase in credit score, given all else being equal.

- c) Using a probability cut-off of 0.5, the model produces the following confusion matrix on the 300 customers (rows = actual, columns = predicted):

	Predicted default	Predicted no default	Total
Actual default	72	18	90
Actual no default	24	186	210
Total	96	204	300

Compute the sensitivity and the specificity of the classifier.

Solution. Treating “default” as the positive class:

$$\text{sensitivity} = \frac{TP}{TP + FN} = \frac{72}{72 + 18} = \frac{72}{90} = 0.800,$$

$$\text{specificity} = \frac{TN}{TN + FP} = \frac{186}{186 + 24} = \frac{186}{210} = 0.886.$$

sensitivity = 0.800, specificity $\approx$ 0.886

- d) The ROC curve of the full model has an area under the curve (AUC) of 0.842, while a reduced model (with `credit_score` removed) has an AUC of 0.788. Which model discriminates better between defaulters and non-defaulters, and why?

Solution. The **full model** discriminates better: a higher AUC indicates a better ability to rank a randomly chosen defaulter above a randomly chosen non-defaulter across all thresholds. Since $0.842 > 0.788$, keeping `credit_score` improves classification performance.

Cheat sheet

Exploring data. $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$, $s = \sqrt{s^2}$, $\text{IQR} = Q_3 - Q_1$, fences $Q_1 - 1.5 \text{IQR}$, $Q_3 + 1.5 \text{IQR}$.

$$r_{(x,y),n} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x s_y}$$

Probability. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ if independent; $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
 $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$.

$$\mathbb{E}(X) = \sum_i x_i \mathbb{P}(X = x_i) \text{ (discrete)}, \quad \mathbb{E}(X) = \int x f(x) dx \text{ (cont.)},$$

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}, \quad \text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y), \quad Z = \frac{X - \mathbb{E}(X)}{\sqrt{\text{Var}(X)}}.$$

Distributions.

$$\text{Geometric: } \mathbb{P}(X = x) = (1-p)^{x-1}p, \quad \mathbb{E}(X) = \frac{1}{p}, \quad \text{SD}(X) = \sqrt{\frac{1-p}{p^2}}.$$

$$\text{Binomial: } \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \mathbb{E}(X) = np, \quad \text{SD}(X) = \sqrt{np(1-p)}.$$

$$\text{Neg. Binomial: } \mathbb{P}(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad x \geq k. \quad \text{Poisson: } \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \mathbb{E}(X) = \lambda, \quad \text{SD}(X) = \sqrt{\lambda}.$$

Inference — categorical.

$$1 \text{ prop. CI: } \hat{p} \pm z^* \text{SE}(\hat{p}), \quad \text{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}; \quad \text{test: } Z = \frac{\hat{p} - p_0}{\text{SE}_0}, \quad \text{SE}_0 = \sqrt{\frac{p_0(1-p_0)}{n}}.$$

$$2 \text{ prop. CI: } (\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}; \quad \text{test: } \text{SE}_0 = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}, \quad \hat{p}_{\text{pool}} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}.$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected}_{ij} = \frac{(\text{row } i)(\text{col } j)}{\text{total}}, \quad df = k - 1 \text{ (GOF) or } (r-1)(c-1).$$

Inference — numerical.

$$1\text{-sample } t: \bar{x} \pm t_{df}^* \text{SE}(\bar{x}), \quad T_{df} = \frac{\bar{x} - \mu_0}{\text{SE}(\bar{x})}, \quad \text{SE}(\bar{x}) = \frac{s}{\sqrt{n}}, \quad df = n - 1.$$

$$2\text{-sample } t: (\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \text{SE}, \quad T_{df} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\text{SE}}, \quad \text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \quad df = \min(n_1 - 1, n_2 - 1).$$

$$\text{Sample size: two-sample } n \geq \frac{2\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2; \quad \text{one-sample } n \geq \frac{\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2, \quad \alpha^* \in \{\alpha, \alpha/2\}.$$

$$\text{ANOVA \& regression. } F = \frac{\text{MSG}}{\text{MSE}}, \quad df_1 = k - 1, \quad df_2 = n - k.$$

$$\text{SSG} = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2, \quad \text{SST} = \sum_{j=1}^n (y_j - \bar{y})^2, \quad \text{SSE} = \text{SST} - \text{SSG}.$$

$$\text{Post-hoc } T_{df} = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\text{MSE}/n_i + \text{MSE}/n_j}}, \quad df = n - k; \quad \text{Bonferroni } \alpha^* = \alpha/K, \quad p_{\text{adj},i} = K p_i, \quad K = \binom{k}{2}.$$

$$\hat{y} = b_0 + b_1 x, \quad b_1 = r_{(x,y),n} \frac{s_y}{s_x}, \quad \hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}, \quad T_{df} = \frac{b_1 - \beta_{1,0}}{\text{SE}_{b_1}}, \quad df = n - 2.$$

$$R^2 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad R_{\text{adj}}^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum (y_i - \bar{y})^2 / (n - 1)}, \quad \text{AIC} = c + n \ln(\sum e_i^2) + 2k.$$

Logistic regression. $\text{logit}(p_i) = \log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}$, $\text{OR}_j = e^{\beta_j}$, sensitivity = $\mathbb{P}(\text{Test+} | \text{Cond+})$, specificity = $\mathbb{P}(\text{Test-} | \text{Cond-})$.

Quantile hints (standard normal). $z_{0.70} = 0.524$, $z_{0.80} = 0.842$, $z_{0.90} = 1.282$, $z_{0.95} = 1.645$, $z_{0.975} = 1.960$, $z_{0.99} = 2.326$, $z_{0.995} = 2.576$.